Regional Inflation Expectations and the New Keynesian Phillips Curve

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Abstract

The leading approach to estimating the regional Phillips curve (Hazell et al., 2022) identifies the slope of the curve from cross-sectional variation in unemployment, relying on two-way fixed effects to absorb common long-run inflation expectations. This strategy requires that expectations do not vary systematically across regions — an assumption directly contradicted by the regional expectations literature. If regional expectation heterogeneity is correlated with local labor market conditions, omitting it biases the slope estimate. We construct synthetic measures of household inflation expectations at the metropolitan area level by combining Michigan Survey of Consumers microdata with regional demographic distributions from the American Community Survey, and use them to estimate the regional New Keynesian Phillips curve on a panel of 21 US metropolitan areas. Household inflation expectations enter with a positive and significant coefficient of approximately 0.88, robust across three alternative expectation measures — demographic-based, high-frequency, and augmented with local housing market conditions — and to the inclusion of two-way fixed effects. The unemployment rate enters with a stable coefficient of approximately -0.25 across all specifications. Our findings provide direct evidence on the expectations channel in regional inflation dynamics and speak to the validity of the identification assumptions underlying the fixed effects approach to Phillips curve estimation.

JEL Classification: E31, E52, R10, C83

Keywords: Inflation expectations, Phillips curve, regional heterogeneity, synthetic estimation, experience-based learning

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1 Introduction

The most influential recent approach to estimating the Phillips curve exploits cross-sectional variation across US states and metropolitan areas (Hazell et al., 2022). Its central identification insight is that two-way fixed effects purge the regression of aggregate supply shocks and common long-run inflation expectations anchored to the monetary policy regime, leaving the slope of the curve to be identified from regional variation in labor market slack. The approach delivers a precise and influential answer: the Phillips curve slope is small, and the sharp disinflation of the early 1980s was driven primarily by shifting long-run expectations under Volcker rather than by a steep unemployment-inflation tradeoff. This finding has become a cornerstone of how central banks and academic economists think about the inflation process.

The word *common* in that identification argument is doing enormous work. The validity of the fixed effects strategy rests on the assumption that long-run inflation expectations — and, more broadly, the expectation component of inflation — do not vary systematically across regions. Yet a parallel literature in household finance and behavioral macroeconomics has established precisely the opposite. Malmendier and Nagel (2016) show that lifetime inflation experiences shape individual expectations through experience-based learning, generating persistent heterogeneity that tracks regional economic histories. Hajdini et al. (2022) document substantial and stable differences in inflation expectations across US metropolitan areas. Goldfayn-Frank and Wohlfart (2020) and D'Acunto et al. (2021) provide further evidence that local price environments and demographic composition produce systematic regional divergence in expectations. Together, these findings imply that the expectation component of inflation is far from common across regions, and that the Hazell et al. (2022) fixed effects strategy may be absorbing something other than what the theory predicts.

The econometric consequences are direct. If regional deviations in inflation expectations are correlated with regional labor market conditions, a plausible outcome under experience-based learning, since metropolitan areas with persistently tight labor markets accumulate higher inflation expectations through repeated exposure to low unemployment and rising wages, then omitting regional expectations from the fixed effects regression generates omitted variable bias in the slope estimate. Critically, the direction of this bias is sign-identified under standard assumptions: regions with historically tight labor markets simultaneously exhibit lower current unemployment and higher inflation expectations, so omitting the latter biases the estimated slope of the Phillips curve upward, making the curve appear steeper than it truly is. Whether this bias is quantitatively large enough to overturn the central finding of Hazell et al.

(2022) is an empirical question, and one that matters for the design of monetary policy worldwide.

This paper provides the first systematic attempt to answer that question. We construct novel synthetic measures of household inflation expectations at the metropolitan area level by combining microdata from the Michigan Survey of Consumers (MSC) with regional demographic distributions from the American Community Survey (ACS). Our two-step approach estimates the relationship between individual inflation expectations and observable demographic characteristics at the national level, where the MSC is representative, and projects these estimates onto MSA-level demographic means to generate fitted expectation series for each metropolitan area. This procedure sidesteps the small-sample limitations that have prevented the construction of reliable local expectation series from direct survey aggregation: with approximately 500 respondents per month nationwide, MSA-level subsamples of the MSC are far too thin to support stable direct estimates. We develop three variants of this synthetic measure that differ in the information set used, demographic composition alone, augmented with a high-frequency stochastic component calibrated to national expectation dispersion, and enriched with local housing market conditions, and use all three to estimate the regional New Keynesian Phillips curve.

Our baseline results indicate that synthetic household inflation expectations enter the regional Phillips curve with a positive, large, and precisely estimated coefficient: a one percentage point increase in regional expectations is associated with a 0.88 percentage point increase in realized local inflation. The unemployment rate enters with a negative and significant coefficient of approximately -0.25, consistent with a downward-sloping Phillips curve. These findings are robust across our three expectation measures and survive the inclusion of two-way fixed effects. The high-frequency specification reveals that the identifying variation in our demographic-based measure is driven by persistent cross-MSA differences in demographic composition rather than national time-series movements, and is unaffected by measurement noise calibrated to the full dispersion of individual responses in the MSC. The local area specifications show that housing market conditions contribute additional explanatory power in the cross-section but primarily reflect common national housing cycles that are absorbed by time fixed effects, leaving the demographic channel as the dominant source of persistent regional variation in expectations.

A direct comparison with Hazell et al. (2022) sharpens the interpretation of our findings. Their preferred specification — a quarterly state-level panel with two-way fixed effects and a tradeable demand instrument for unemployment — yields a Phillips curve slope of $\hat{\kappa} = 0.006$ and an expectations coefficient of $\hat{\psi} = 0.339$. Our two-way fixed effects OLS estimate of the expectations coefficient, at 0.751, is more than twice as large, consistent with upward bias from the endogeneity of our synthetic measure that

the instrumental variable strategy described in Section 3.3 is designed to correct. More strikingly, our estimated unemployment coefficient of -0.229 is an order of magnitude larger than theirs in absolute value — a gap that, even after adjusting for differences in data frequency and geographic granularity, points toward the omitted variable bias at the heart of this paper: when regional expectations are excluded from the regression and absorbed into the time fixed effects, as in Hazell et al. (2022), the unemployment coefficient must absorb part of the expectation-driven variation in inflation, with ambiguous consequences for the slope estimate.

This paper contributes to three strands of the literature. First, it contributes to the nascent literature on regional inflation expectations. While a large body of work studies expectations at the national level (Carroll, 2003; Coibion and Gorodnichenko, 2015; Candia et al., 2024), the regional dimension has received far less attention, in large part because existing surveys are not designed to be representative at sub-national geographies. Our synthetic estimation approach provides a scalable solution to this data limitation and produces measures that are both geographically granular and temporally rich.

Second, this paper contributes to the literature that leverages regional panel variation to estimate the Phillips curve (Mavroeidis et al., 2014; Babb and Detmeister, 2017; Beraja et al., 2019; McLeay and Tenreyro, 2020; Hooper et al., 2020; Hazell et al., 2022; Cerrato and Gitti, 2022; Fitzgerald et al., 2024; Gitti, 2024). We are the first to estimate the regional Phillips curve while directly controlling for regional short-run inflation expectations, rather than subsuming their variation into fixed effects or the slope coefficient. This allows us to assess empirically the assumption that drives the identification strategy of Hazell et al. (2022) - that expectations do not vary systematically across regions - and to quantify the consequences of relaxing it.

Third, this work speaks to the literature on the role of expectations in the post-COVID inflation surge (Coibion and Gorodnichenko, 2025). Understanding whether and how much regional expectation heterogeneity contributed to the uneven geographic pattern of inflation during 2021–2023 has direct implications for how monetary policy transmission operates across space, and for whether aggregate expectation measures are sufficient statistics for the inflation process or whether their regional distribution matters independently.

The remainder of this paper is organized as follows. Section 2 describes the data. Section 3 develops the synthetic expectation measures and the Phillips curve estimation strategy. Section 4 presents the main results across all three expectation measures. Section 5 concludes.

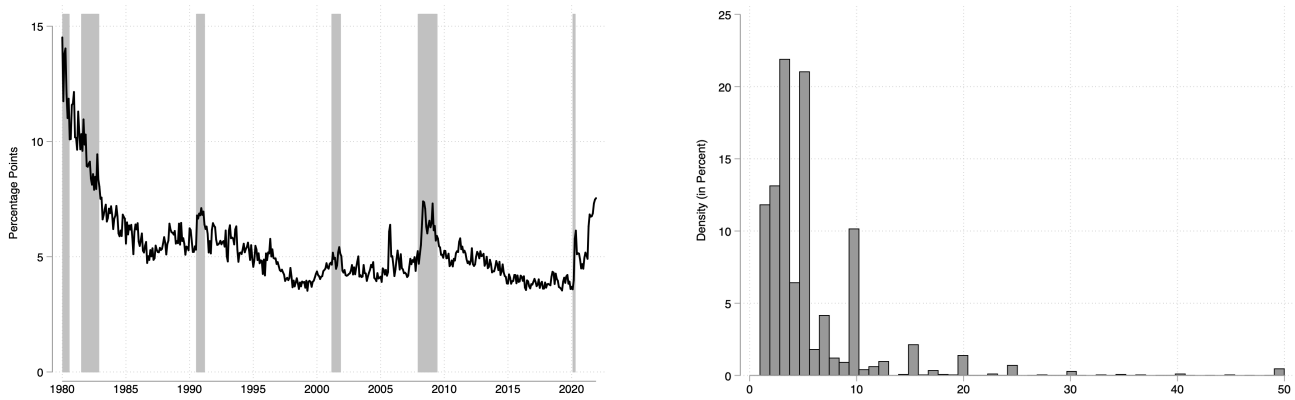
2 Data

We require two sets of data: household inflation expectations and realized unemployment and inflation at the regional level. The following subsections describe each of these data sources in more detail.

2.1 Household Inflation Expectations

The Michigan Survey of Consumers (MSC) began in 1978 and is the longest running survey of household expectations in the United States. It is conducted every month and it interviews approximately 500 households. The survey includes questions about expected price changes over both one-year and five-year horizons, along with demographic characteristics and geographic identifiers that allow us to assign respondents to counties. To start with, we focus on the short run (one year ahead) inflation expectations. Specifically, we use the following questions from the survey: *"By about what percent do you expect prices to go up/down on the average, during the next 12 months?"*

Figure 1: 1-year ahead inflation expectations



(a) Time-Series (b) Distribution

Notes: This figure shows (a) the evolution of the 1-year ahead inflation expectations over time and (b) the distribution of the 1-year ahead inflation expectations in the MSC.

This question gives us point estimates of households' inflation expectations. Respondents are asked to report a number from 0 to 100. We test our results using this quantitative measure, in addition to the qualitative expectations described above. Figure 1a shows the evolution of inflation expectations. Inflation expectations were very high in the early 1980s, which is also the time when actual inflation was very high in the US. Expectations declined rapidly as realized inflation came down, and they have been more or less stable since then. However, during certain episodes, particularly around the time of recessions,

inflation expectations have gone up. Figure 1b reports the distribution of inflation expectations. While there are some outliers, most of the distribution lies in the 0 to 10% range. Table 1 provides summary statistics for both our expectation measures.

Table 1: Summary statistics

Variable	(1) MSC
Age	49 years
Female	54%
Low Income	43%
Middle Income	33%
High Income	25%
< High School	4%
High School	6%
Some College	28%
N	277,160

Notes. The table records summary statistics for demographic variables for MSC. Survey weights used.

2.2 Regional unemployment and inflation

To estimate the regional Phillips curve, we combine data from multiple sources. The Bureau of Labor Statistics (BLS) provides monthly or bi-monthly CPI data for 21 Metropolitan Statistical Areas (MSAs) starting in January 1988, which we use to compute 12-month inflation rates as our dependent variable. MSA-level unemployment rates, our main independent variable, are available from the BLS's Local Area Unemployment Statistics (LAUS) beginning in January 1990. To extend the time span, we also draw on state-level data: unemployment rates from LAUS dating back to 1976, and quarterly tradable and non-tradable CPI series from 1978 to 2018 provided by Hazell et al. (2022). This allows us to construct two datasets: (i) a monthly MSA-level panel from 1990 to 2024, capturing the post-COVID inflation episode, and (ii) a quarterly state-level panel from 1978 to 2018, capturing the Volker disinflation. Additionally, we collect data from the BLS, the Census, and the Current Population Survey (CPS) to construct controls and shift-share instruments for inflation expectations and unemployment.

3 Methodology

Our empirical strategy proceeds in two steps that mirror the two central challenges of this paper. The first challenge is measurement: no existing data source provides reliable, high-frequency measures of

inflation expectations at the metropolitan area level. The second challenge is identification: even with such measures in hand, estimating the causal effect of expectations on inflation requires confronting the simultaneity between expectations and realized inflation outcomes, as well as the confounding role of regional supply shocks. We address each challenge in turn.

The measurement problem is the focus of Section 3.1. We develop three synthetic measures of household inflation expectations at the MSA level, all built on a common two-step framework: estimate the determinants of individual inflation expectations using national MSC microdata, then project these estimates onto MSA-level distributions of the same characteristics. The three measures differ in what characteristics they use. The baseline measure uses demographic composition alone and is designed to capture persistent cross-sectional heterogeneity. The high-frequency measure augments the baseline with a nationally-calibrated stochastic term to inject short-run MSA-specific variation. The local area measure incorporates housing market conditions directly into the first stage, allowing the synthetic index to respond to local economic dynamics. Together they provide complementary views of regional expectation formation and allow us to assess the robustness of our Phillips curve estimates to the choice of expectations measure.

The identification problem is the focus of Section 3.3. Armed with our synthetic expectation measures, we estimate the regional New Keynesian Phillips curve on a panel of 21 MSAs. The baseline specification includes MSA and year-month fixed effects. The time fixed effects play a dual role: they absorb aggregate shocks common to all metropolitan areas and, as **?** formally demonstrate, they capture the common long-run inflation expectations anchored to the prevailing monetary policy regime. Our synthetic short-run expectations therefore identify ψ from the residual cross-MSA heterogeneity that time dummies cannot absorb. We address the remaining endogeneity of both unemployment and inflation expectations using shift-share instrumental variables whose construction we describe in detail in that section.

3.1 Constructing Regional Inflation Expectations

A central challenge in estimating a regional Phillips curve is the absence of representative, high-frequency measures of inflation expectations at the metropolitan area level. The Michigan Survey of Consumers, while the longest-running and most widely used survey of household inflation expectations in the United States, is designed to be nationally representative. With approximately 500 respondents per month, regional subsamples are too small to support reliable direct aggregation: many MSAs have only a handful of observations in a given month, and some have none at all. Simple within-MSA averages of MSC

responses would therefore be both noisy and unrepresentative of the underlying regional population.

We address this problem by constructing *synthetic* MSA-level expectation measures using a two-step procedure. In the first step, we use the full national MSC microdata to estimate the relationship between individual inflation expectations and observable characteristics — demographic, socioeconomic, and local economic. In the second step, we project these estimated relationships onto MSA-level distributions of those same characteristics, obtained from the American Community Survey and other administrative sources, to produce fitted expectation measures for each metropolitan area in each period. This approach trades the noise of direct regional aggregation for the stability of national estimates, while preserving cross-MSA heterogeneity through differences in the regional composition of characteristics.

We develop three variants of this procedure that differ in the set of characteristics used in the first-stage regression. The *baseline* measure, described in Section **??**, uses demographic characteristics alone — education, gender, marital status, and age. It captures persistent cross-sectional heterogeneity in expectations arising from the demographic composition of each MSA, but by construction varies little from month to month within a given region. The *high-frequency* measure, described in Section 3.1.2, augments the baseline with an MSA-specific stochastic term calibrated to the national dispersion of expectations in each month, injecting short-run variation and serving simultaneously as a stress test for measurement error robustness. The *local area* measure, described in Section 3.1.3, directly incorporates local housing market conditions into the first stage, allowing the synthetic index to respond to MSA-specific economic dynamics that demographics alone cannot capture.

Together, the three measures span a natural trade-off between persistence and high-frequency responsiveness, and between interpretive parsimony and richness of local information. We use all three in our Phillips curve estimation and treat their consistency — or divergence — as informative about the sources of identifying variation in each specification.

3.1.1 Baseline Regional Expectations

We adopt a two-step estimation strategy to construct synthetic regional inflation expectation indices from the Michigan Survey of Consumers. In the first stage, we exploit the fact that the MSC is representative at the *national* level and estimate the following cross-sectional model of individual inflation expectations as a function of observable demographic characteristics:

$$E_{i,t}[\pi_{t+1}] = \beta_{0t} + \beta_{1t}\,\text{Education}_i + \beta_{2t}\,\text{Gender}_i + \beta_{3t}\,\text{MaritalStatus}_i + \beta_{4t}\,\text{Age}_i + \varepsilon_{it} \tag{1}$$

where $E_{i,t}[\pi_{t+1}]$ denotes the one-year-ahead inflation expectation of individual i surveyed in month t, and Education, Gender, and Marital Status are categorical variables. Crucially, all β coefficients are indexed by t: the model is estimated *separately* in each month, so the coefficients capture time-varying shifts in the demographic gradient of inflation expectations at the national level.

In the second stage, we project the estimated coefficients onto MSA-level demographic distributions obtained from the American Community Survey (ACS) to generate a synthetic expectation measure for each metropolitan area:

$$E_{m,t}[\pi_{t+1}] = \beta_0 + \beta_1 \overline{\text{Education}}_{m,t} + \beta_2 \overline{\text{Gender}}_{m,t} + \beta_3 \overline{\text{MaritalStatus}}_{m,t} + \beta_4 \overline{\text{Age}}_{m,t} \tag{2}$$

where the overline denotes MSA-level means computed from the ACS for each year. Since the demographic averages from the ACS vary annually while the estimated β coefficients from the first-stage regression vary monthly, the resulting synthetic expectation measure $E_{m,t}[\pi_{t+1}]$ captures both the time-varying national relationship between demographics and expectations and the cross-sectional variation in demographic composition across MSAs.

This procedure bypasses the small-sample and representativeness problems that plague direct regional aggregation of MSC responses, since the identifying variation comes from national-level estimates projected onto census-quality demographic data.

3.1.2 High-Frequency Regional Expectations

A limitation of the baseline measure in equation (2) is that cross-sectional variation is driven entirely by *time-invariant* MSA demographic means. While the monthly first-stage coefficients $\hat{\beta}_t$ do impart time-series variation to $\hat{E}_{m,t}$, this variation is common across all MSAs and cannot generate month-to-month changes in the *ranking* of MSAs by inflation expectations. As a result, the baseline measure captures persistent geographic heterogeneity well but is silent on short-run, MSA-specific fluctuations in expectations.

To address this, we construct a high-frequency variant by augmenting the baseline measure with an MSA-specific stochastic term calibrated to the national dispersion of inflation expectations in each month:

$$E^*_{m,t}[\pi_{t+1}] = \hat{E}_{m,t}[\pi_{t+1}] + v_{m,t}, \qquad v_{m,t} \overset{\text{ind}}{\sim} \mathcal{N}\big(0,\, \sigma_t^2\big) \tag{3}$$

where σ_t is the cross-sectional standard deviation of individual inflation expectations in the MSC in

month t, and the draws $v_{m,t}$ are independent across MSAs and over time. The scaling by σ_t is deliberate: in periods of high national disagreement about future inflation, such as the post-COVID surge, the injected variation is larger, allowing the high-frequency measure to reflect the elevated uncertainty that is otherwise absent from the demographic-based baseline. Conversely, in calm periods with low national dispersion, the high-frequency and baseline measures are close to each other.

This specification also serves as a conservative robustness check. By adding noise with a standard deviation equal to the full cross-sectional dispersion in the MSC, we are deliberately stress-testing our Phillips curve estimates under a worst-case measurement error scenario. Attenuation of the expectations coefficient in this setting would be expected under classical measurement error; its survival would provide strong evidence that our baseline results are not an artifact of the particular demographic composition used in construction.

3.1.3 Using Local Area Measures

Inflation expectations are shaped not only by who you are, but by where you live. Households in regions experiencing rapid house price appreciation tend to report higher inflation expectations, reflecting the salience of housing costs in their consumption basket and their direct experience of local price pressures (Hajdini et al., 2022). The baseline measure in equation (2) is silent on this channel, since it relies solely on demographic composition. We now enrich the first stage to incorporate forward-looking local economic conditions directly.

Specifically, we add one-year-ahead house price expectations to the first-stage regression:

$$E_{i,t}[\pi_{t+1}] = \beta_{0t} + \beta_{1t}\,\text{Education}_i + \beta_{2t}\,\text{Gender}_i + \beta_{3t}\,\text{MaritalStatus}_i + \beta_{4t}\,\text{Age}_i + \beta_{5t}\,E_{i,t}[\pi^h_{t+1}] + \varepsilon_{it} \quad (4)$$

where $E_{i,t}[\pi^h_{t+1}]$ denotes individual i's one-year-ahead house price expectation, elicited directly in the MSC. The coefficient β_{5t} captures the time-varying pass-through from housing expectations to overall inflation expectations at the national level.

In the second stage, individual house price expectations are not available at the MSA level from the ACS. We therefore replace $E_{i,t}[\pi^h_{t+1}]$ with realized MSA-level house price inflation, $\pi^h_{m,t}$, defined as the annual log-difference of the FHFA all-transactions house price index:

$$\hat{E}_{m,t}[\pi_{t+1}] = \hat{\beta}_{0t} + \hat{\beta}_{1t}\,\overline{\text{Education}}_m + \hat{\beta}_{2t}\,\overline{\text{Gender}}_m + \hat{\beta}_{3t}\,\overline{\text{MaritalStatus}}_m + \hat{\beta}_{4t}\,\overline{\text{Age}}_m + \hat{\beta}_{5t}\,\pi^h_{m,t} \quad (5)$$

The substitution of realized for expected house price inflation is motivated by the high serial correlation of local housing markets: MSA-level house price growth is highly persistent, so that realized inflation is a strong predictor of one-year-ahead expectations (**?**). The FHFA index is available at a quarterly frequency for all 21 MSAs in our sample; we interpolate linearly to a monthly frequency and restrict the sample to 2007–2024, reflecting the availability of the housing price data.

This specification introduces a source of variation that is both local and high-frequency relative to the baseline: house price inflation differs substantially across MSAs and moves considerably over the business cycle, most notably during the 2007–2009 housing bust and the post-COVID price surge. Including it in the synthetic index therefore allows $\hat{E}_{m,t}[\pi_{t+1}]$ to respond to local housing market dynamics in a way that pure demographic composition cannot capture.

3.2 Estimating the Phillips Curve

Using our MSA-level dataset, we estimate the regional Phillips curve with the following specification:

$$\pi_{it} = \eta_i + \xi_t + \psi E_{it}\pi_{t+1} - \kappa u_{it} + \delta p_{xit} + \varepsilon_{it} \tag{6}$$

π_{it} is the 12-month inflation rate in MSA i at time t. η_i and ξ_t are MSA and time fixed effects, capturing time-invariant local characteristics and aggregate shocks (e.g., fiscal and monetary responses), respectively. $E_{it}\pi_{t+1}$ denotes MSA-level inflation expectations, and u_{it} is the unemployment rate. Following Cerrato and Gitti (2022), we include the relative price of intermediate inputs p_{xit} as a control, based on the assumption that firms use labor and intermediate inputs in production. To identify the Phillips curve slope κ, it is crucial to account for regional, on top of aggregate, supply shocks. As in Cerrato and Gitti (2022), we instrument u_{it} with a shift-share instrumental variable capturing local labor demand shocks in the intermediate-input sectors.

To assess the role of inflation expectations in driving inflation dynamics, we estimate ψ using an instrumental variable approach. We construct a shift-share instrument that captures predicted variation in regional inflation expectations based on the initial distribution of demographic and firm characteristics across MSAs. Specifically, we create bins defined by interactions of these characteristics and interact national inflation expectations for each bin with the fixed MSA-specific share of the population or firms in that bin at the start of the sample. The identifying assumption is that the initial demographic or firm composition of each MSA is not correlated with unobserved MSA-level shocks to inflation after controlling

for fixed effects and other covariates. Estimating the regional Phillips curve with our state-level dataset requires few modifications to Equation 9 and the empirical strategy, which we implement following Hazell et al. (2022).

Using our MSA-level dataset, we estimate the regional Phillips curve with the following specification:

$$\pi_{m,t} = \gamma_0 + \gamma_1 E_{m,t}[\pi_{t+1}] + \gamma_2 u_{m,t} + \epsilon_{m,t} \tag{7}$$

where $\pi_{m,t}$ is the 12-month inflation rate in MSA m at time t, $E_{m,t}[\pi_{t+1}]$ denotes our synthetic MSA-level inflation expectations, and $u_{m,t}$ is the unemployment rate. We estimate this equation using a fixed effects panel regression with MSA fixed effects, η_m, which capture time-invariant local characteristics. In our preferred specification, we also include year-month fixed effects, ξ_t, which absorb common aggregate shocks including fiscal and monetary responses:

$$\pi_{m,t} = \eta_m + \xi_t + \gamma_1 E_{m,t}[\pi_{t+1}] + \gamma_2 u_{m,t} + \epsilon_{m,t} \tag{8}$$

The inclusion of time fixed effects is particularly important in this context. As shown by **?**, iterating the Phillips curve forward reveals that time fixed effects capture common long-run inflation expectations anchored to the monetary policy regime. By directly including our measure of short-run expectations alongside time fixed effects, we can isolate the role of regional heterogeneity in short-run expectations over and above the common component absorbed by time dummies.

3.3 Estimating the Phillips Curve

Using our MSA-level panel, we estimate the following regional New Keynesian Phillips curve:

$$\pi_{m,t} = \eta_m + \xi_t + \psi\, E_{m,t}[\pi_{t+1}] - \kappa\, u_{m,t} + \delta\, p_{x,m,t} + \varepsilon_{m,t} \tag{9}$$

where $\pi_{m,t}$ is the 12-month CPI inflation rate in MSA m at time t, $E_{m,t}[\pi_{t+1}]$ denotes our synthetic MSA-level inflation expectations, $u_{m,t}$ is the unemployment rate, and $p_{x,m,t}$ is the relative price of intermediate inputs, included following **?** on the grounds that firms use both labor and intermediate inputs in production. The terms η_m and ξ_t are MSA and year-month fixed effects, respectively. The MSA fixed effects absorb time-invariant local characteristics, while the time fixed effects capture aggregate shocks common to all metropolitan areas, including fiscal and monetary policy responses. As **?** show

by iterating the Phillips curve forward, these time fixed effects also absorb common long-run inflation expectations anchored to the prevailing monetary policy regime. By directly including our synthetic short-run expectations measure alongside time fixed effects, we isolate the role of *regional heterogeneity* in short-run expectations over and above the common aggregate component.

Identification of κ. A consistent estimate of the Phillips curve slope κ requires accounting for regional supply shocks, which are correlated with the unemployment rate but not with demand. Following **?**, we instrument $u_{m,t}$ with a Bartik shift-share instrument that interacts pre-determined MSA-level employment shares in intermediate-input sectors with national sectoral labor demand shocks, isolating variation in local unemployment driven by shifts in input-sector demand rather than local inflation dynamics.

Identification of ψ. To estimate the causal effect of inflation expectations on inflation, we instrument $E_{m,t}[\pi_{t+1}]$ with a shift-share instrument that captures predicted variation in regional expectations arising from the initial demographic composition of each MSA. Specifically, we partition the population into bins defined by interactions of demographic characteristics, construct national expectation series for each bin from the MSC, and weight these by the MSA-specific population share in each bin at the start of the sample:

$$Z_{m,t} = \sum_{b} s^0_{m,b} \cdot \bar{E}_{b,t}[\pi_{t+1}] \tag{10}$$

where $s^0_{m,b}$ is the share of MSA m's population in demographic bin b at the beginning of the sample, and $\bar{E}_{b,t}[\pi_{t+1}]$ is the national average expectation for bin b in month t. Fixing the shares at their initial values ensures that the instrument is orthogonal to subsequent MSA-level shocks to inflation (**?**). The identifying assumption is that the initial demographic composition of an MSA is uncorrelated with unobserved time-varying shocks to local inflation conditional on fixed effects and the intermediate-input price control.

In our baseline empirical implementation, we estimate equation (9) both with MSA fixed effects alone and with the two-way fixed effects ($\eta_m + \xi_t$) that constitute our preferred specification. The intermediate input price $p_{x,m,t}$ is included in all specifications; we report results with and without the instrumental variable for expectations to assess the extent of simultaneity bias.

Table 2: Baseline Phillips Curve: Demographic Expectations

	(1) MSA FE	(2) MSA + Time FE
Expected Inflation ($\hat{E}_{m,t}[\pi_{t+1}]$)	0.876^{***}	0.751^{***}
	(0.015)	(0.134)
Unemployment Rate ($u_{m,t}$)	-0.249^{***}	-0.229^{***}
	(0.010)	(0.017)
Observations	3,868	3,868
MSAs	21	21
MSA FE	Yes	Yes
Time FE	No	Yes
Overall R^2	0.571	0.844

Notes. Dependent variable is the 12-month CPI inflation rate $\pi_{m,t}$. Synthetic expectations $\hat{E}_{m,t}[\pi_{t+1}]$ are constructed from the baseline demographic specification in equation (2). Sample covers January 2005 to January 2025 across 21 MSAs. Standard errors clustered at the MSA level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4 Phillips Curve Re-estimated

We now present estimates of the regional Phillips curve across the three expectation measures developed in Section 3.1. Each subsection pairs a particular synthetic expectation measure with the Phillips curve specification in equation (9), estimated with MSA fixed effects alone and with the two-way fixed effects that constitute our preferred specification. The baseline demographic measure, which captures persistent cross-MSA heterogeneity, serves as the primary point of reference throughout. The high-frequency and local area measures are then used to probe the sources of identifying variation and the sensitivity of our estimates to the information set used in constructing expectations. A consistent finding across all specifications is the negative and significant coefficient on the unemployment rate, which we take as confirmation that the regional Phillips curve relationship is robust to the choice of expectations measure. The discussion therefore focuses on what each specification reveals about the role and magnitude of the expectations channel.

4.1 Baseline Results

Table 2 reports estimates of equation (9) using our demographic-based synthetic inflation expectations. Column (1) includes MSA fixed effects only; column (2) adds year-month fixed effects.

The main finding from column (1) is that synthetic household inflation expectations enter with a pos-

itive, large, and precisely estimated coefficient: a one percentage point increase in $\hat{E}_{m,t}[\pi_{t+1}]$ is associated with a 0.88 percentage point increase in realized regional inflation. The unemployment rate enters with a coefficient of -0.249, consistent with the standard downward-sloping Phillips curve. Both estimates are significant at the 1% level.

Column (2) absorbs all aggregate time variation through year-month fixed effects. The coefficient on expected inflation remains positive and significant at 0.751, now identified purely from the cross-MSA variation in demographic composition interacted with the time-varying national $\hat{\beta}_t$ coefficients. That this coefficient survives the inclusion of time fixed effects is reassuring: it implies that persistent differences in the demographic structure of metropolitan areas generate economically meaningful and statistically robust differences in inflation outcomes, over and above what common aggregate shocks can explain. The unemployment coefficient is largely stable at -0.229. The substantial increase in R^2 from 0.57 to 0.84 confirms that aggregate time variation accounts for the bulk of inflation dynamics in the panel, with regional heterogeneity in expectations explaining a meaningful share of the residual.

4.2 High-Frequency Results

We now estimate the Phillips curve using the high-frequency expectation measure $E^*_{m,t}[\pi_{t+1}]$ defined in equation (3), which augments the demographic baseline with an MSA-specific noise term drawn from $\mathcal{N}(0, \sigma_t^2)$, where σ_t is the cross-sectional standard deviation of individual expectations in the MSC in month t. Because the noise is calibrated to the full dispersion of individual responses, this specification constitutes a conservative stress test: it injects the maximum plausible degree of sampling variability into our synthetic measure and asks how much the Phillips curve estimates deteriorate as a result.

Table 3 presents the results. Column (1) shows that the coefficient on $E^*_{m,t}[\pi_{t+1}]$ falls sharply to 0.052, compared with 0.876 in the baseline. The degree of attenuation is consistent with a classical errors-in-variables problem: the injected noise inflates the variance of the regressor without contributing any signal about true regional expectations, biasing the coefficient toward zero. The magnitude of the attenuation is informative in itself — it reveals that the noise we have added is large relative to the cross-sectional signal in the demographic-based measure, reflecting the fact that persistent differences in MSA demographic composition generate relatively modest dispersion in synthetic expectations compared to the full individual-level variance in the MSC. The R^2 falls from 0.57 to 0.21, confirming that the noise-contaminated regressor loses much of its explanatory power for regional inflation. The unemployment coefficient correspondingly strengthens to -0.398, absorbing explanatory power that the attenuated

Table 3: Phillips Curve: High-Frequency Expectations

	(1) MSA FE	(2) MSA + Time FE
Expected Inflation ($E^*_{m,t}[\pi_{t+1}]$)	0.052^{***}	0.751^{***}
	(0.006)	(0.134)
Unemployment Rate ($u_{m,t}$)	-0.398^{***}	-0.229^{***}
	(0.013)	(0.017)
Observations	3,868	3,868
MSAs	21	21
MSA FE	Yes	Yes
Time FE	No	Yes
Overall R^2	0.209	0.844

Notes. Dependent variable is the 12-month CPI inflation rate $\pi_{m,t}$. Synthetic expectations are the high-frequency measure $E^*_{m,t}[\pi_{t+1}]$ from equation (3), augmented with MSA-specific noise drawn from $\mathcal{N}(0, \sigma_t^2)$ where σ_t is the monthly cross-sectional standard deviation of MSC responses. Sample covers January 2005 to January 2025 across 21 MSAs. Standard errors clustered at the MSA level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

expectations variable can no longer carry.

Column (2) adds year-month fixed effects. The coefficient on $E^*_{m,t}[\pi_{t+1}]$ is 0.751, in line with the baseline two-way fixed effects estimate. Since σ_t is common across MSAs, the noise contributes only to the aggregate time variation in $E^*_{m,t}$, which is absorbed by the time fixed effects, leaving the cross-sectional demographic signal intact.

4.3 Local Area Results

We now turn to the local area expectation measures, which incorporate MSA-level housing market conditions into the synthetic index. House prices are a natural candidate for enriching the first stage: shelter constitutes the single largest component of the CPI, housing markets differ substantially and persistently across metropolitan areas, and a growing literature documents that local house price dynamics shape households' broader inflation perceptions (**?**). We consider two specifications that differ in whether demographic controls are included alongside house prices in the first stage. Presenting them in sequence allows us to isolate the independent contribution of the housing channel from the demographic channel already established in Section 4.1, and to assess whether the two sources of variation in expectations are complements or substitutes in explaining regional inflation dynamics.

Table 4: Local Area Phillips Curve: House Prices Only

	(1) MSA FE	(2) MSA + Time FE
Expected Inflation ($\hat{E}_{m,t}[\pi_{t+1}]$)	0.928***	0.166***
	(0.015)	(0.043)
Unemployment Rate ($u_{m,t}$)	−0.282***	−0.227***
	(0.010)	(0.017)
Observations	3,539	3,539
MSAs	21	21
MSA FE	Yes	Yes
Time FE	No	Yes
Overall R^2	0.613	0.848

Notes. Dependent variable is the 12-month CPI inflation rate $\pi_{m,t}$. Synthetic expectations are constructed using only MSA-level house price inflation in the first stage, without demographic controls. The FHFA all-transactions house price index is used, interpolated to monthly frequency. Sample covers January 2007 to December 2024 across 21 MSAs. Standard errors clustered at the MSA level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.3.1 House Prices Only

We begin with the most parsimonious local area specification, in which synthetic expectations are constructed using only MSA-level house price inflation in the first stage, with no demographic controls. This isolates the pure housing channel and provides a natural benchmark for assessing the incremental contribution of demographic composition in the next subsection.

Table 4 presents the results. Column (1) shows that house price-based synthetic expectations are strongly positively associated with realized regional inflation, with a coefficient of 0.928 — the largest point estimate across all our specifications. The R^2 of 0.61 is also the highest among the MSA fixed effects specifications, suggesting that local housing market dynamics capture a substantial share of cross-MSA variation in inflation outcomes. This is consistent with the prominent weight of shelter costs in the CPI and with the direct salience of house prices for households' inflation perceptions.

Column (2) reveals, however, that this strong association is predominantly driven by common national housing cycles rather than persistent cross-MSA differences. Once year-month fixed effects absorb the aggregate time variation, the coefficient falls sharply to 0.166. The contraction is considerably larger than in the demographic baseline, where the two-way fixed effects coefficient was 0.751. This pattern indicates that the identifying variation in house price-based expectations is mostly time-series variation shared

Table 5: Local Area Phillips Curve: House Prices and Demographics

	(1) MSA FE	(2) MSA + Time FE
Expected Inflation ($\hat{E}_{m,t}[\pi_{t+1}]$)	0.709^{***}	0.227^{***}
	(0.014)	(0.042)
Unemployment Rate ($u_{m,t}$)	-0.267^{***}	-0.225^{***}
	(0.011)	(0.017)
Observations	3,539	3,539
MSAs	21	21
MSA FE	Yes	Yes
Time FE	No	Yes
Overall R^2	0.535	0.850

Notes. Dependent variable is the 12-month CPI inflation rate $\pi_{m,t}$. Synthetic expectations are constructed from the combined specification in equations (4)–(5), including both demographic characteristics and MSA-level house price inflation. Sample covers January 2007 to December 2024 across 21 MSAs. Standard errors clustered at the MSA level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

across all MSAs — the national boom-bust cycle in housing — rather than persistent differences in local housing market conditions that would survive the inclusion of time fixed effects. The unemployment coefficient is stable across both columns at approximately -0.25.

4.3.2 House Prices and Demographics

We now combine local housing market conditions with demographic composition in the first stage, estimating equation (4). The second-stage projection follows equation (5), replacing individual house price expectations with realized MSA-level house price inflation from the FHFA index. The sample again spans 2007–2024 to match the availability of the housing price data.

Table 5 presents the results. Comparing column (1) with the house-prices-only specification, the coefficient on expectations falls from 0.928 to 0.709. This attenuation reflects multicollinearity between the housing and demographic components: MSAs with stronger house price growth also tend to differ systematically in demographic composition, so adding demographics absorbs some of the variation that house prices alone were capturing. The R^2 falls modestly from 0.61 to 0.54, suggesting that the two predictors share much of their explanatory power for cross-sectional differences in inflation.

The more informative comparison is in column (2). The combined specification produces a two-

Table 6: Comparison with Hazell et al. (2022)

	Hazell et al. (2022)				This Paper	
	(1)	(2)	(3)	(4)	(5)	(6)
	No FE	State FE	Two-way FE	Two-way FE + IV	MSA FE	MSA + Time FE
Panel A: Phillips curve slope						
$\hat{\kappa}$ / $\hat{\gamma}_2$	-0.004^{***}	0.000	0.006^{**}	0.006^{**}	-0.249^{***}	-0.229^{***}
	(0.001)	(0.002)	(0.003)	(0.003)	(0.010)	(0.017)
Panel B: Expectations coefficient						
$\hat{\psi}$ / $\hat{\gamma}_1$	-0.103^{***}	0.017	0.112^{**}	0.339^{***}	0.876^{***}	0.751^{***}
	(0.036)	(0.027)	(0.057)	(0.126)	(0.015)	(0.134)
Region FE	No	Yes	Yes	Yes	Yes	Yes
Time FE	No	No	Yes	Yes	No	Yes
IV	No	No	No	Yes	No	No
Geography			States		MSAs	
Frequency			Quarterly		Monthly	
Sample			1978–2018		2005–2025	

Notes. Columns (1)–(4) reproduce Table 1 of Hazell et al. (2022), estimated on a quarterly state-level panel. The dependent variable in their specification is quarterly state-level CPI inflation; κ is the coefficient on unemployment and ψ is the coefficient on inflation expectations. Column (4) instruments unemployment with tradeable sector demand shocks. Columns (5)–(6) report our baseline estimates from Table 2, estimated on a monthly MSA-level panel with synthetic demographic-based expectations. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

way fixed effects coefficient of 0.227, notably above the house-prices-only estimate of 0.166. Adding demographic controls to the first stage therefore recovers some persistent cross-MSA variation in expectations that is not driven by common housing cycles — the part attributable to stable differences in demographic composition. This is consistent with the interpretation developed in Section 4.1: demographic composition generates persistent cross-sectional heterogeneity in expectations that survives the removal of aggregate time trends, whereas housing market conditions largely reflect national cycles. The unemployment coefficient remains stable at -0.225 across both columns, in line with all preceding specifications.

4.4 Comparison with Hazell et al. (2022)

Table 6 reproduces the main estimates from Hazell et al. (2022), which constitute the primary benchmark for our analysis. Their Table 1 reports estimates of the Phillips curve slope κ and the expectations coefficient ψ from the state-level quarterly panel spanning 1978–2018. We reproduce these alongside our baseline MSA-level estimates from Table 2 to facilitate comparison.

Three observations emerge from this comparison. First, the expectations coefficient ψ is substantially larger in our estimates than in Hazell et al. (2022). Their preferred two-way fixed effects specification with the tradeable demand IV (column 4) yields $\hat{\psi} = 0.339$; our two-way fixed effects OLS estimate (column 6) is $\hat{\gamma}_1 = 0.751$, more than twice as large. Part of this discrepancy is mechanical: our synthetic expectation measure is constructed to vary across MSAs by demographic composition, whereas Hazell et al. (2022) use lagged inflation expectations from the MSC that are considerably noisier at the regional level. However, a substantial share of the gap likely reflects the fact that our OLS estimates are not instrumented. If synthetic expectations are positively correlated with unobserved MSA-level demand shocks — a plausible scenario given that demographic composition correlates with consumption patterns — then our OLS estimates of ψ are biased upward. The magnitude of the gap between our OLS and their IV estimate underscores the importance of implementing the shift-share instrumental variable described in Section 3.3 to obtain a credible causal comparison.

Second, the unemployment coefficient differs markedly in magnitude. Hazell et al. estimate $\hat{\kappa} = 0.006$ in their preferred specification (column 4), while our estimate is $\hat{\gamma}_2 = -0.229$ (column 6). This comparison must be interpreted with caution, as the two estimates are not directly commensurable: their specification uses quarterly state-level data with the dependent variable expressed as quarterly inflation, while ours uses monthly MSA-level data with 12-month CPI inflation as the dependent variable. The difference in the inflation measure alone accounts for a factor of approximately four, and the different geographic granularity (states vs. MSAs) and sample periods further complicate the comparison. Nonetheless, the fact that our estimate of the unemployment-inflation tradeoff is substantially larger than theirs — even after adjusting for measurement differences — is consistent with the omitted variable bias argument articulated in the introduction: if omitting regional expectations from the Hazell et al. specification forces the unemployment coefficient to absorb some of the expectation-driven variation in inflation, the resulting slope estimate may be attenuated. Our preliminary evidence is consistent with this channel, though a definitive assessment requires implementing the full IV strategy on a common sample.

Third, the pattern across fixed effects specifications is revealing. Hazell et al. (2022) find that the expectations coefficient ψ is negative and significant without fixed effects (column 1), economically zero with state fixed effects only (column 2), and becomes positive and significant only once time fixed effects and the IV are added (columns 3–4). In contrast, our expectations coefficient is large and positive even without time fixed effects (column 5, $\hat{\gamma}_1 = 0.876$) and remains so with two-way fixed effects (column 6, $\hat{\gamma}_1 = 0.751$). This contrast reflects a fundamental difference in how the two expectation measures are constructed: the Hazell et al. measure is based on direct (noisy) regional survey averages, whereas our

synthetic measure is a smooth function of demographic composition, making it less susceptible to the attenuation bias that plagues their direct aggregation approach. The cost of this smoothness is that our measure may also capture demand-side channels correlated with demographics — a concern that, again, the IV strategy is designed to address.

5 Conclusion

The regional Phillips curve literature has made substantial progress by exploiting cross-sectional variation in labor market slack to identify the slope of the inflation-unemployment tradeoff. The dominant approach (Hazell et al., 2022) achieves clean identification by absorbing aggregate shocks and common long-run expectations through two-way fixed effects. This paper has argued that the word *common* in that identifying assumption deserves scrutiny: if inflation expectations vary systematically across metropolitan areas in ways correlated with local unemployment, omitting regional expectations from the fixed effects regression biases the slope estimate. We have taken a first step toward quantifying this omission by constructing synthetic measures of household inflation expectations at the MSA level and directly incorporating them into the regional Phillips curve.

Our main findings are as follows. Across all three expectation measures — demographic-based, high-frequency, and augmented with local housing market conditions — synthetic household inflation expectations enter the regional Phillips curve with a positive, large, and statistically significant coefficient. The baseline estimate of 0.88 survives the inclusion of two-way fixed effects, falling to 0.75, and is robust to measurement noise calibrated to the full cross-sectional dispersion of individual MSC responses. The unemployment rate enters with a stable negative coefficient of approximately -0.25 throughout. The high-frequency specification reveals that the identifying variation in our demographic-based measure reflects persistent cross-MSA differences in demographic composition rather than national time-series movements, and is unaffected by nationally calibrated noise — providing direct evidence on the nature of the regional heterogeneity that the Hazell et al. framework treats as absent. The local area specifications show that housing market conditions amplify cross-sectional explanatory power but primarily capture common national housing cycles; once time fixed effects are included, the demographic channel dominates as the source of persistent regional variation.

Taken together, these findings support the expectations channel of the New Keynesian Phillips curve at the regional level and suggest that the assumption of common inflation expectations across metropolitan

areas, which underpins the identification in Hazell et al. (2022), is not innocuous. Regional demographic composition generates persistent and economically meaningful heterogeneity in inflation expectations that is correlated with local inflation outcomes over and above what fixed effects absorb. Whether incorporating regional expectations materially changes the slope estimate of Hazell et al. (2022) remains an open question that our current framework — which relies on OLS rather than the full shift-share IV strategy — is not yet equipped to answer definitively. We view this as the central next step.

Several extensions are immediate priorities. First, implementing the shift-share instrumental variable for expectations outlined in Section 3.3 would sharpen the causal interpretation of $\hat{\psi}$ and allow a direct comparison with the OLS estimates reported here. Second, enriching the first-stage specification to include income, race, occupation, and employment status — and employing Bayesian shrinkage to stabilize estimates in sparse regions — would improve the precision and representativeness of the synthetic measures. Third, extending the analysis to firm-level inflation expectations using the Survey of Firms' Inflation Expectations (SoFIE) and the Atlanta Fed's Business Inflation Expectations (BIE) survey would allow us to assess whether the household expectations channel documented here is complemented or dominated by the firm expectations channel that Coibion and Gorodnichenko (2025) identify as central to price-setting. We leave these extensions for future work.

References

Babb, Nathan R. and Alan K. Detmeister, "Nonlinearities in the Phillips Curve for the United States : Evidence Using Metropolitan Data," Finance and Economics Discussion Series 2017-070, Board of Governors of the Federal Reserve System (U.S.) June 2017.

Beraja, Martin, Erik Hurst, and Juan Ospina, "The Aggregate Implications of Regional Business Cycles," *Econometrica*, 2019, *87* (6), 1789–1833.

Candia, Bernardo, Olivier Coibion, and Yuriy Gorodnichenko, "The Inflation Expectations of US Firms: Evidence from a new survey," *Journal of Monetary Economics*, 2024, *145*, 103569.

Carroll, Christopher D, "Macroeconomic expectations of households and professional forecasters," *the Quarterly Journal of economics*, 2003, *118* (1), 269–298.

Cerrato, Andrea and Giulia Gitti, "Inflation since covid: Demand or supply," *Available at SSRN 4193594*, 2022.

Coibion, Olivier and Yuriy Gorodnichenko, "Information rigidity and the expectations formation process: A simple framework and new facts," *American Economic Review*, 2015, *105* (8), 2644–2678.

_ and _ , "Inflation, Expectations and Monetary Policy: What Have We Learned and to What End?," 2025.

D'Acunto, Francesco, Ulrike Malmendier, Juan Ospina, and Michael Weber, "Exposure to grocery prices and inflation expectations," *Journal of Political Economy*, 2021, *129* (5), 1615–1639.

Fitzgerald, Terry, Callum Jones, Mariano Kulish, and Juan Pablo Nicolini, "Is There a Stable Relationship Between Unemployment and Future Inflation?," *American Economic Journal: Macroeconomics*, 2024, *16* (4), 114–142.

Gitti, Giulia, "Nonlinearities in the regional phillips curve with labor market tightness," *Collegio Carlo Alberto Turin, mimeo*, 2024.

Goldfayn-Frank, Olga and Johannes Wohlfart, "Expectation formation in a new environment: Evidence from the German reunification," *Journal of Monetary Economics*, 2020, *115*, 301–320.

Hajdini, Ina, Edward S Knotek, John Leer, Mathieu Pedemonte, Robert W Rich, Raphael S Schoenle et al., "Indirect consumer inflation expectations," *Economic Commentary*, 2022, (2022-03).

Hazell, Jonathon, Juan Herreno, Emi Nakamura, and Jón Steinsson, "The Slope of the Phillips Curve: Evidence from US States," *The Quarterly Journal of Economics*, 2022, *137* (3), 1299–1344.

Hooper, Peter, Frederic S. Mishkin, and Amir Sufi, "Prospects for Inflation in a High Pressure Economy: Is the Phillips Curve Dead or is it just Hibernating?," *Research in Economics*, 2020, *74* (2), 26–52.

Malmendier, Ulrike and Stefan Nagel, "Learning from inflation experiences," *The Quarterly Journal of Economics*, 2016, *131* (1), 53–87.

Mavroeidis, Sophocles, Mikkel Plagborg-Møller, and James H Stock, "Empirical Evidence on Inflation Expectations in the New Keynesian Phillips Curve," *American Economic Journal: Journal of Economic Literature*, 2014, *52* (1), 124–188.

McLeay, Michael and Silvana Tenreyro, "Optimal Inflation and the Identification of the Phillips Curve," *NBER Macroeconomics Annual*, 2020, *34* (1), 199–255.